AeroGard: Hyperlocal Air-Quality Forecasting, Exposure Estimation and Explainable Respiratory-Risk Alerts for Asthma and COPD

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***Abstract*—People with asthma or chronic obstructive pulmonary disease (COPD) usually learn about air pollution from a city-wide index that describes the present, not the air around them tomorrow, and says nothing about their own exposure or health. We built AeroGard, a software prototype that forecasts air quality 24 hours ahead at a user's location, estimates personal exposure, and turns these with a symptom diary into a calibrated, explained personal risk and an alert. For Bengaluru we ingested 753,481 hourly observations from 14 regulatory monitors (February 2025 to September 2026) with reanalysis weather. A direct multi-horizon LightGBM forecaster reached a 24-hour $PM_{2.5}$ mean absolute error of 5.78 µg/m³ on an untouched test period, 23% better than persistence. Because no respiratory outcome data exist for the city, the risk layer was developed on a clearly labelled synthetic cohort of 400 patients simulated inside the real pollution field. A LightGBM model with a shared trunk and asthma and COPD heads, Platt-calibrated, reached an AUROC of 0.708 (95% CI 0.684–0.732) with an expected calibration error of 0.0025, and 0.721 for patients never seen in training. The symptom diary carried the signal; estimated exposure added no measurable discrimination. Exact TreeSHAP explanations, scaled to add up to the probability shown, and an idempotent alert engine are served through an authenticated, consent-aware API. These are synthetic-data results, not clinical validation.**

***Index Terms*—air quality forecasting, PM2.5, LightGBM, exposure assessment, asthma, COPD, risk prediction, probability calibration, SHAP, synthetic cohort, data leakage**

I. Introduction

Air pollution is one of the leading environmental risks to health worldwide [1]. In India it was linked to about 1.67 million deaths in 2019 [2]. Short-term rises in particulate matter, nitrogen dioxide and ozone are associated with higher mortality [3] and with more asthma emergency visits and hospital admissions [4], so people living with asthma or COPD are among those who most need timely and relevant warnings.

In practice most people check a city-level Air Quality Index. That number describes the city now, not the neighbourhood tomorrow, and it ignores where a person spends their time and how sensitive they are. Building even a reliable picture of the present is harder than it looks: monitoring stations are sparse, public feeds arrive late, readings drop out, and, as we found, units can be mislabelled.

Here we describe AeroGard, a system that forecasts air quality for each of the next 24 hours at a user's location, estimates personal exposure from location and microenvironment, and converts this, together with a symptom diary, into a calibrated personal risk, an explanation and an alert. The contributions are: (1) a validated ingestion pipeline for Bengaluru's regulatory monitors, including the detection and correction of a unit error in the public feed; (2) a leakage-audited 24-hour forecaster compared on equal terms with two baselines; (3) a transparent exposure estimate; (4) a synthetic asthma and COPD cohort simulated inside the real pollution field, on which a calibrated risk model is compared with baselines and ablations under a protocol fixed before testing; and (5) exact, additive explanations and an idempotent alert engine served through a secured API. We report negative findings as such: on this cohort, pollution exposure did not improve risk discrimination.

II. Problem Statement and Objectives

Generic air-quality monitoring reports environmental conditions but does not answer the question a patient actually has: given the air expected around me, my recent exposure and my health, what is my respiratory risk over the next 24 hours? There is a need for a system that produces location-specific forecasts, estimates exposure honestly, and communicates risk without making clinical claims.

Our objectives are to (1) collect and validate hourly pollutant and weather data for Bengaluru; (2) match a user's location to nearby monitors; (3) forecast pollutant concentrations for each of the next 24 hours and evaluate the forecasts against observed data; (4) estimate personal exposure from location and activity context; (5) predict calibrated asthma and COPD risk and map it to alert levels; and (6) deliver the results through a secure API and an Android application. Objectives (1)–(5) and the API are complete; the Android application is in progress.

III. Literature Review

Table I lists ten studies we reviewed, with the method and data each used, its main finding and its main limitation.

TABLE I
SUMMARY OF RELATED WORK

Ref.	Year	Method / Data	Key Finding	Limitation
[2]	2021	Burden-of-disease estimates for Indian states	Air pollution linked to about 1.67 M deaths in India in 2019	Population level; no individual risk
[3]	2020	Meta-analysis of short-term exposure studies	PM, NO_2 and O_3 associated with higher mortality	Pooled effects; not a prediction tool
[4]	2015	Meta-analysis of time-series studies	Pollutants associated with asthma ER visits and admissions	Aggregate outcomes, not personal
[5]	2012	Review of real-time AQ forecasting	Surveys chemical-transport and statistical approaches	Predates current ML methods
[6]	2017	LSTM with spatio-temporal inputs, Beijing	Outperformed statistical baselines for $PM_{2.5}$	One city; data-hungry model
[7]	2012	Multi-step forecasting strategies, NN5 data	Direct and multi-output strategies beat recursive	Not specific to air quality
[8]	2004	Interpolation of monitor data (IDW, kriging, others)	Method choice changes exposure estimates	Depends on monitor density
[9]	2011	Review of indoor/outdoor particle relationships	I/O ratios vary with building and ventilation	Few South-Asian buildings
[10]	2022	Connected inhaler, peak-flow and wearable data + ML	Framework for predicting asthma attacks	Protocol stage; no AQ forecast
[11]	2014	Review of air pollution in Indian cities	Documents sources and sparse monitoring	Descriptive; no forecasting

Epidemiological work establishes that short-term exposure matters [1]–[4], but its results are population averages rather than personal predictions. Forecasting research ranges from chemical-transport models to data-driven methods [5]; deep sequence models can outperform statistical baselines [6], and the choice between recursive and direct multi-step strategies affects accuracy [7]. Exposure studies show that an outdoor monitor is an imperfect proxy for what a person breathes: the interpolation method [8] and indoor infiltration [9] both change the estimate. Connected-device asthma studies [10] predict attacks from physiological and inhaler data but do not use air-quality forecasts. Finally, results in this area are easy to inflate through data leakage, where information unavailable at prediction time enters the model [12].

Research gap. Existing work treats air-quality forecasting, exposure estimation and respiratory risk separately. We found few systems for Indian cities that combine station-level 24-hour forecasts, a transparent exposure estimate and personal risk in one traceable pipeline, evaluated with explicit leakage checks against simple baselines.

IV. PROPOSED METHODOLOGY

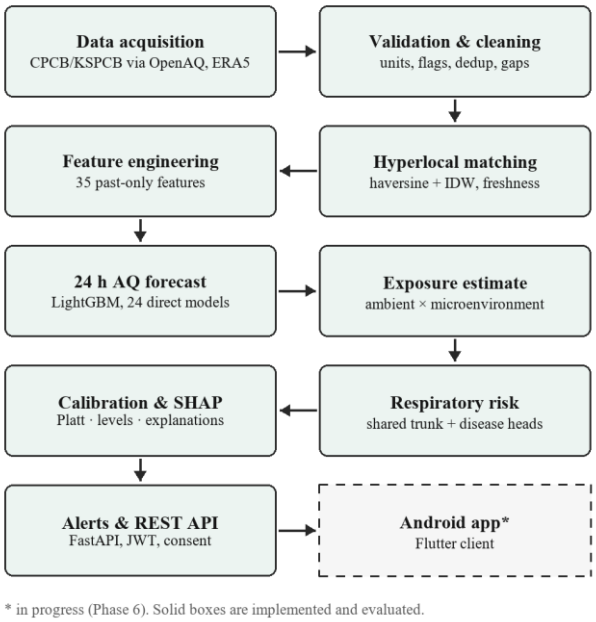


Fig. 1. Methodology pipeline. The dashed stage is in progress.

Fig. 1 shows the pipeline. Hourly pollutant means from regulatory monitors are obtained through the OpenAQ API [13], and hourly ERA5 reanalysis weather [14] through Open-Meteo [15]. Records are validated and cleaned, matched to the user's location, turned into features and passed to the 24-hour forecaster. The forecast and the user's location history drive the exposure estimate, which, together with the health profile and symptom diary, feeds the respiratory-risk model, its calibration, the explanation and the alert engine.

Validation and cleaning. Every record is converted to a canonical form: a UTC timestamp at the end of the averaging hour, a pollutant identity, and a concentration in µg/m³. Gases reported in ppb are converted at 25 °C and 1 atm (µg/m³ = ppb × M / 24.45, with M the molecular weight). Records with missing timestamps, unknown pollutants, invalid coordinates, or missing, non-finite or negative values are rejected and the reason is counted. Hours with fewer than 75% of their sub-hourly readings, provider flags or physically implausible values are kept but flagged. Short gaps of up to 3 h are forward-filled from past values only, and outliers are flagged with a trailing robust z-score (median and MAD over the previous 24 h) rather than deleted.

Hyperlocal matching. For a location we find reference monitors within 10 km by haversine distance, discard readings older than 3 h or later than the query time, and combine up to four stations by inverse-distance weighting with power 2 [8], halving the weight of low-coverage readings. If no usable station remains, the system reports the value as unavailable instead of extrapolating.

V. DATASET

The study area is Bengaluru, India. A 25 km search around the city centre returned 14 CPCB/KSPCB reference monitors

with data in the study period; low-cost sensors listed on the same platform were excluded from ground truth. All stations share a gap from about October 2022 to February 2025, when the source feed changed, so we use 18 February 2025 to 21 September 2026, about 19 months. Hourly weather (13,992 h) comes from ERA5 at the city centre. Table II summarises the pollutant data; about 0.3% of fetched records were rejected, all for missing values.

TABLE II
DATASET COMPOSITION (BENGALURU, FEB 2025 – SEP 2026)

Pollutant	Stations	Hourly rows	Flagged	Median (µg/m³)
SO_2	14	133,264	13.1%	19.0
PM_{10}	14	131,429	11.7%	68.7
NO_2	14	130,878	12.0%	32.9
CO	14	129,348	15.0%	600.0
O_3	12	114,773	12.5%	23.7
$PM_{2.5}$	12	113,789	12.3%	25.3

The current feed labels CO, NO_2 and SO_2 in ppb. For CO this cannot be right: at BTM Layout the median reading was 0.49 in June–September 2025 against 1,260 µg/m³ on the older feed in the same months of 2021, and ambient CO is typically several hundred ppb. The values match CPCB's reporting unit for CO, mg/m³ [16], so all CO rows were converted accordingly. For NO_2 and SO_2 the comparison is ambiguous; their absolute levels are therefore treated as unverified and are not used in alert thresholds. This does not affect forecasting, because tree-based models are unchanged by a constant rescaling of an input.

VI. MODEL DEVELOPMENT

Each training row is a (station, origin hour t) pair with 35 features: the current value of the target pollutant and whether it was filled; lags of 1, 6, 12, 24 and 48 h; the 1-h difference; rolling mean, maximum and standard deviation over 6 h and 24 h; the number of observed hours in the last day; the current values of the other pollutants; weather at t (temperature, humidity, wind speed and direction as sine and cosine, pressure, precipitation and its 24-h total); calendar terms in Indian Standard Time; and station coordinates. Every feature declares how far back it reads, and an automated test confirms that altering any data after t leaves the features at t unchanged, guarding against leakage [12].

We use a direct multi-horizon strategy [7]: one LightGBM gradient-boosted tree model [17] per horizon h = 1, …, 24, each trained on log(1 + y) and back-transformed, with negative outputs clipped to zero. No forecast is ever fed back in as an input. We chose LightGBM because it trains quickly on a CPU, handles missing values natively, performs well on tabular data and produces small artifacts that are easy to deploy. Two baselines complete the comparison: persistence, $\hat{y}(t+h) = y(t)$, and seasonal naive, $\hat{y}(t+h) = y(t+h-24)$.

Targets are observed values from hours flagged valid. Inputs may still use flagged hours, but isolated $PM_{2.5}$ readings of 600–900 µg/m³ during the monsoon, mostly at 25–50% coverage, looked like sensor faults and were not used as ground truth. Results on all observed hours are reported as a secondary check.

VII. EXPERIMENTAL SETUP

Table III gives the training configuration and the environment used for the experiments.

TABLE III
EXPERIMENTAL SETUP

Item	Value
Split (chronological)	Train 18 Feb 2025 – 31 Jan 2026; validation 1 Feb – 30 Apr 2026; test 1 May – 21 Sep 2026
Origins ($PM_{2.5}$)	Train 72,342; validation 18,576; test 22,396
Embargo	24 h before each boundary, so no target crosses a split
Model	LightGBM, 24 direct models per pollutant
Hyperparameters	≤ 600 trees, learning rate 0.03, 31 leaves, min. 50 samples per leaf, row and column subsampling 0.8, L2 = 1
Early stopping	50 rounds on validation (best 65–233 trees)
Target transform	log(1 + y)
Evaluation	Test period used once; all models scored on identical pairs
Software	Python 3.14, LightGBM 4.7, scikit-learn 1.9, pandas 3.0, PostgreSQL 16
Hardware	Consumer laptop CPU, Windows 11; no GPU

VIII. EVALUATION METRICS

We report mean absolute error (MAE), root mean squared error (RMSE), the coefficient of determination (R^2), mean bias, and skill relative to persistence. With $\hat{y}_i$ the forecast, y_i the observation and n the number of forecast–observation pairs,

$$MAE = (1/n)\ \Sigma\ |\hat{y}_i - y_i| \quad (1)$$

$$RMSE = \sqrt{[(1/n)\ \Sigma\ (\hat{y}_i - y_i)^2]} \quad (2)$$

$$R^2 = 1 - \Sigma\ (\hat{y}_i - y_i)^2 / \Sigma\ (y_i - \bar{y})^2 \quad (3)$$

$$Skill = 1 - MAE_{model} / MAE_{persistence} \quad (4)$$

where $\bar{y}$ is the mean observation. Metrics are computed per horizon and pooled over all 24 horizons. To keep the comparison fair, every model is scored on the same (origin, horizon) pairs, namely those where all models produce a forecast and a valid observation exists; otherwise a baseline that cannot forecast difficult cases would look better than it is.

IX. FORECAST RESULTS

Table IV gives pooled test-set performance for four pollutants and Table V the $PM_{2.5}$ error by horizon. The test period covers the monsoon season.

TABLE IV
TEST SET PERFORMANCE (VALID HOURS)

Pollutant	Model	MAE (µg/m³)	RMSE (µg/m³)	R^2	Skill
$PM_{2.5}$	LightGBM	5.78	15.67	0.31	+23%
$PM_{2.5}$	Seasonal naive	6.92	22.96	−0.49	+8%
$PM_{2.5}$	Persistence	7.51	22.82	−0.47	+0%
PM_{10}	LightGBM	14.01	26.12	0.56	+26%
PM_{10}	Seasonal naive	15.61	34.34	0.24	+18%
PM_{10}	Persistence	19.03	38.80	0.03	+0%
NO_2	LightGBM	5.74	37.69	0.24	+28%
NO_2	Seasonal naive	6.83	45.79	−0.13	+14%
NO_2	Persistence	7.93	45.39	−0.11	+0%

Pollutant	Model	MAE (µg/m³)	RMSE (µg/m³)	R^2	Skill
O_3	Seasonal naive	4.83	11.32	0.52	+28%
O_3	LightGBM	5.02	10.51	0.59	+25%
O_3	Persistence	6.68	14.53	0.21	+0%

TABLE V
$PM_{2.5}$ MAE BY FORECAST HORIZON

Model (MAE, µg/m³)	1 h	3 h	6 h	12 h	18 h	24 h
LightGBM	3.18	4.99	5.71	5.97	6.12	6.39
Seasonal naive	7.24	7.01	6.87	6.85	6.88	7.02
Persistence	3.32	6.08	7.66	8.47	8.18	7.02

LightGBM had the lowest MAE for $PM_{2.5}$, PM_{10} and NO_2, improving on persistence by 23%, 26% and 28%. Table V shows where the gain comes from: at 1 h ahead persistence is almost as good, but its error grows quickly with horizon, whereas LightGBM's grows slowly and stays below the seasonal baseline at every horizon. The same ordering held on the validation period (February–April, the dry season, when $PM_{2.5}$ is higher): MAE 12.63 for LightGBM against 14.38 for seasonal naive and 16.98 for persistence.

Three results qualify this picture. For O_3, the seasonal-naive baseline has the lowest MAE, although LightGBM has the lower RMSE and higher R^2; ozone follows the daily cycle of sunlight so closely that yesterday's value at the same hour is hard to beat. For NO_2 at 1 h ahead, persistence beats LightGBM. For $PM_{2.5}$, RMSE is well above MAE and R^2 is modest, which means the model improves typical hours but does not anticipate sudden spikes. All forecasts are slightly biased low ($PM_{2.5}$ bias −1.60 µg/m³), as expected from training on a log scale. Scoring against all observed hours, including flagged ones, gives the same conclusion ($PM_{2.5}$ MAE 5.95 against 7.67 for persistence).

X. EXPOSURE ESTIMATION

Exposure is an estimate, not a measurement: no personal sensor is used. For each hour we take the ambient concentration at the user's location from the matching in Section IV and multiply it by a factor for the user's microenvironment (Table VI). The factors are central assumptions drawn from indoor/outdoor studies [9], not values calibrated to AeroGard users.

TABLE VI
MICROENVIRONMENT FACTORS (EXPOSURE / AMBIENT)

Setting	$PM_{2.5}$	PM_{10}	NO_2	O_3	CO
Outdoors / unknown	1.0	1.0	1.0	1.0	1.0
Indoors, home	0.8	0.6	0.6	0.3	0.9
Indoors, other	0.6	0.5	0.5	0.2	0.8
In traffic	1.3	1.3	1.6	0.8	1.8

If no location fix exists in the two hours before a given hour, that hour's exposure is reported as missing rather than guessed, and an unknown microenvironment is treated as outdoors and flagged. Six-, 24- and 72-hour means, maxima and cumulative exposure (µg/m³·h) are reported together with their coverage and marked incomplete below 75%. Because the factors are assumptions, the risk results were re-run with the factors scaled and with the exact simulated exposure (Section XIV).

XI. SYSTEM ARCHITECTURE AND IMPLEMENTATION

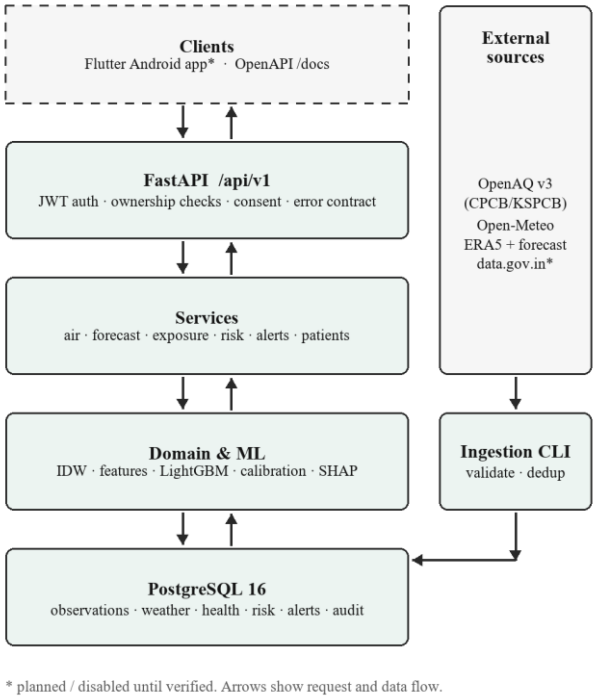


Fig. 2. System architecture of the platform.

Fig. 2 shows the architecture. Clients call a versioned REST API built with FastAPI. Routes stay thin and delegate to services, which use repositories for database access and the domain and machine-learning modules for matching, cleaning, features and inference. An ingestion command-line tool pulls data from the external sources, validates it and writes it to PostgreSQL.

Backend. The API exposes current, historical and forecast air quality for a location, reference stations, location upload, exposure, symptom-diary records, the current and 24-hour rolling risk with its explanation, risk history, and alerts with acknowledgement. Models come from frozen, versioned files; each carries a metadata record with its training period, metrics and checksum, and the service refuses to load a file whose checksum or feature list does not match the code. If no model is available the API returns an explicit error rather than a placeholder value.

Security and privacy. Passwords are hashed with Argon2 and sessions use short-lived JSON Web Tokens with rotating refresh tokens. Every patient-scoped operation checks ownership, so one user cannot read another user's data, and location and exposure require recorded consent. Sensitive actions are written to an audit log that stores counts and field names, never health or location values.

Testing. An automated suite of 401 tests runs against a separate database built through the real migrations. It covers

authentication and token misuse, patient isolation, source adapters under malformed input, timeouts and partial failure, validation and unit conversion, idempotent reloading, spatial edge cases, leakage guards for both feature sets, forecast and risk horizons, probability bounds, calibration, explanation additivity and consistency, alert rules and deduplication, and artifact checks.

XII. Synthetic Respiratory Cohort

No respiratory outcome data exist for Bengaluru, and collecting them requires ethics approval and a follow-up far longer than this project. The risk layer was therefore developed on a simulated panel of patients. The air-quality and weather inputs are real; the patients, their routines, exposure, symptoms and exacerbations are simulated. Every risk result in this paper is a synthetic-data evaluation, not clinical validation.

The cohort has 400 patients (250 asthma, 150 COPD) with severity, age, home and workplace drawn from configured distributions, hidden frailty and pollution sensitivity, and daily routines (home, commute, work, errands, walks). True personal exposure is the real station field interpolated to each place, perturbed for street-scale variation and scaled by a personal microenvironment factor. The system only sees what production would see: location logged with a patient-specific rate and day-long dropouts, an unknown setting 10% of the time, and exposure estimated by the live matching rule. Events follow an hourly hazard,

$$\log \lambda = c_s + u + s\beta\Delta_{2.5} + \beta^T z + \beta_I I + \beta_C C \quad (5)$$

for patient i at hour t (indices omitted), where c_s is solved so each disease × severity stratum has its configured annual rate, u is frailty, s pollution sensitivity, $\Delta_{2.5}$ the 24-h exponential moving average of true $PM_{2.5}$ exposure above 25 µg/m³ per 10 µg/m³, z collects PM_{10}, O_3, cold, humidity, recent-event and nocturnal-asthma terms, I is a seasonal viral infection, the dominant trigger of exacerbations [23], and C is a latent daily loss-of-control state that also drives the diary, standing for the worsening that precedes most exacerbations. A diary at 21:00 reports a 0–10 symptom score, reliever puffs and an exacerbation flag, filled with patient-specific adherence; 40% of patients also wear a device. The pollution effects are of the order reported in short-term exposure studies [3], [4] but were chosen to give a learnable signal and are not calibrated to Indian epidemiology. A scale factor on all pollution terms provides a negative control (zero) and stronger scenarios.

The simulation produced 3,449 events (5.4 per patient-year), 74% diary completeness and 94% location availability. The target is whether an event starts within the next 24 hours of an origin; two random origins are drawn per patient-day, excluding warm-up, the final day and ongoing episodes.

XIII. Risk Model, Calibration, Explanations and Alerts

Each origin has 38 features computed only from data at or before it, with the same code in training and in the API: estimated exposure ($PM_{2.5}$ now, 6-, 24- and 72-h means, 24-h maximum, change and coverage; PM_{10} and O_3 summaries), weather, calendar terms, profile (age, sex, disease, severity) and health history (last and mean symptom scores, trend, reliever use, diary days, events in 90 days, days since the last event, and the last wearable readings). Missing values stay missing, as in the training data, so the model learns a branch for them. A test perturbs every input after the origin and requires identical features.

The model is a shared LightGBM trunk trained on all patients with disease as a feature, plus one small LightGBM head per disease boosted from the trunk's output,

$$p = \sigma(m_{trunk}(x) + m_d(x)) \quad (6)$$

where σ is the logistic function and early stopping on validation keeps a head near zero if the disease adds nothing. Baselines use the same protocol: the base rate of the patient's stratum, logistic regression with missing-value indicators, one pooled LightGBM, and one LightGBM per disease; two ablations remove the exposure and the health features. Of the seen patients (80%), origins up to 31 January 2026 train (201,422 rows, 2,974 events), February–April validate (52,623, 832) and May–September test (84,908, 1,241); the other 20% of patients are tested over the same months (21,293, 311). Origins whose label window crosses a boundary are dropped, the protocol was fixed before testing, and 95% intervals come from a patient-level cluster bootstrap with 200 resamples.

Probabilities are calibrated per disease on validation with Platt scaling [19],

$$p_{cal} = \sigma(a \cdot \text{logit}\, p + b) \quad (7)$$

after isotonic regression was rejected because its lowest block reported an exact zero risk for 4–6% of test predictions. Discrimination is measured by AUROC and AUPRC, accuracy of the probabilities by the Brier score [20] and the expected calibration error over B = 10 equal-frequency bins [21],

$$ECE = \Sigma_b\, (n_b/n)\, |\bar{o}_b - \bar{p}_b| \quad (8)$$

where $\bar{o}_b$ and $\bar{p}_b$ are the observed rate and mean prediction in bin b. A versioned policy maps the calibrated probability to Low, Moderate, High or Severe per disease, with thresholds fixed on validation: Moderate at 1.5 × the base rate, High at the F_1-optimal alert threshold, and Severe at the top 1% of validation predictions. An optional personal offset shifts the log-odds by the patient's own history,

$$p' = \sigma(\text{logit}\, p_{cal} + \delta) \quad (9)$$

with $\delta = \text{clip}(\ln((O + 2)/(E + 2)), \pm \ln 2)$, O the exacerbations the patient reported before the prediction time and E the number expected from their stratum over the same diary days, so the offset shrinks to zero for new users.

Explanations. Each prediction is explained with exact TreeSHAP [18], [22] computed by LightGBM. SHAP values are additive, so the trunk and head attributions sum, and because Platt scaling is linear in log-odds the attributions can be moved onto the probability the patient sees:

$$\text{logit}\, p' = (a\varphi_0 + b) + \Sigma_j\, a\varphi_j(x) + \delta \quad (10)$$

where φ_0 is the base value and φ_j the attribution of feature j. The 38 features are grouped into 14 patient-facing factors, and the largest five are worded as having raised or lowered the

estimate, never as causes. Each explanation is stored with the prediction's model, feature and calibration versions and a fingerprint of its input vector, and is refused on read if any of them differs.

Alerts. An alert is created when the current estimate is valid, at least High, made no more than 2 h before evaluation, and no alert exists for that prediction or for an equivalent one, meaning a non-cancelled alert at the same or a higher level whose 24-h window is still running. A higher level supersedes open lower alerts; a recomputed prediction that falls below High withdraws its own alert. Predictions and the alert decision are written in one transaction with a row lock on the patient. An alert states only the level and percentage, never medication advice.

XIV. Risk Results

Table VII compares the models on the test period. The test prevalence is 1.5% per 24 hours, so a random model has an AUPRC of about 0.015.

TABLE VII
Risk Model Comparison, Test, Seen Patients (Synthetic Labels)

Model	AUROC [95% CI]	AUPRC	Brier	ECE
Stratum prior (no features)	0.597 [0.564, 0.632]	0.020	0.01438	0.0018
Logistic regression	0.699 [0.674, 0.724]	0.043	0.01424	0.0025
LightGBM pooled	0.709 [0.685, 0.732]	0.043	0.01423	0.0025
LightGBM per disease	0.701 [0.679, 0.724]	0.043	0.01423	0.0022
Shared trunk + heads (served)	0.708 [0.684, 0.732]	0.043	0.01423	0.0025
Ablation: no exposure	0.709 [0.685, 0.732]	0.045	0.01421	0.0025
Ablation: no health	0.583 [0.555, 0.608]	0.020	0.01439	0.0022

The served model reached an AUROC of 0.708 and generalised to unseen patients (0.721). Its expected calibration error was 0.0025; Fig. 3 shows the observed event rate slightly above the prediction in most bins, because the monsoon test period has more infections than the February–April period on which the calibrators were fitted, a seasonal base-rate shift. Three findings should be stated plainly. First, the symptom diary carries the predictive signal: without health features the AUROC falls to 0.583, close to the stratum prior. Second, the shared-head architecture is not measurably better than one pooled LightGBM or logistic regression; all intervals overlap. Third, removing the exposure features does not lower discrimination, at the default effect size or at two and four times it.

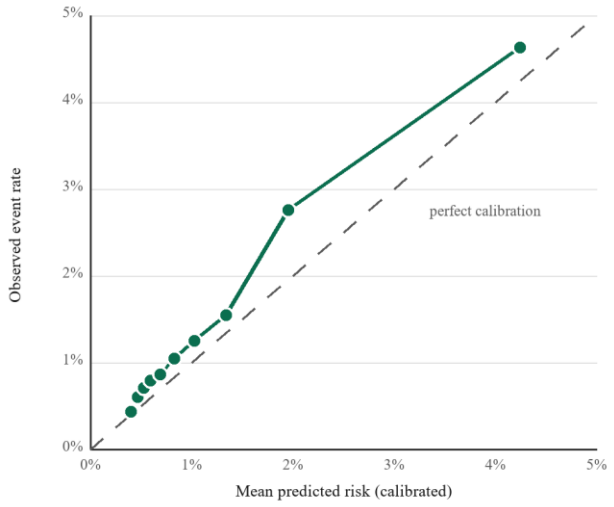


Fig. 3. Reliability of the calibrated risk, test, seen patients (10 equal-frequency bins).

Two reasons explain the third finding. The test period is the monsoon, when exposure is low and narrow ($PM_{2.5}$ averaging 17 µg/m³, mostly below the 25 µg/m³ reference of the hazard), so the pollution term varies little; and the simulated symptom score already responds to pollution, so the diary absorbs most of it. The pre-registered test period was not moved.

TABLE VIII
Risk Levels on Test: Share of Time and Observed Event Rate

Disease	Measure	Low	Moderate	High	Severe
Asthma	Share of time	88.9%	8.2%	2.1%	0.8%
Asthma	Event rate	1.2%	4.2%	7.0%	11.5%
COPD	Share of time	89.8%	8.5%	1.3%	0.5%
COPD	Event rate	0.9%	3.3%	4.4%	5.0%

Event rates rise with every level for both diseases (Table VIII), so the levels separate risk. Table IX compares alerting strategies, with High or above as an alert for the model.

TABLE IX
Alerting Strategies, Test, Seen Patients

Strategy	Precision	Recall	F_1	Time alerting
Generic AQ threshold (NAQI Poor)	0.043	0.001	0.002	0.0%
Disease-specific AQ threshold	0.025	0.011	0.015	0.6%
Model, one global threshold	0.069	0.123	0.089	2.6%
Model, per-disease thresholds	0.072	0.120	0.090	2.5%
Model + personal offset	0.067	0.178	0.097	3.9%

Bengaluru's air almost never crosses the national index thresholds, so alerting on air quality alone is nearly silent. The personal offset raised recall from 0.12 to 0.18 at the cost of alerting more often (2.5% to 3.9% of the time) while leaving probability quality unchanged: a different operating point, not better discrimination. Even at High, most alerts are not followed by an event, which is expected at a 1.5% base rate and is why the level and percentage are both shown.

TABLE X
EXPOSURE SENSITIVITY: 24-H $PM_{2.5}$ EXPOSURE ERROR AND RISK AUROC

Exposure used	MAE (µg/m³)	Bias	Spearman ρ	AUROC
Estimated (production rule)	3.99	−0.21	0.800	0.708
Microenvironment factors × 0.8	4.50	−3.18	0.799	0.708
Microenvironment factors × 1.2	4.97	+2.76	0.800	0.708
Ambient at home only	5.76	+3.66	0.777	0.706
True exposure (oracle)	0.00	+0.00	1.000	0.708

Table X re-trains the model on each exposure variant. The production estimate is the least biased; ignoring mobility raises the error by about 44% and overstates exposure, because people spend time indoors. Even the exact simulated exposure does not change the AUROC, consistent with the finding above.

Fig. 4 shows the global importance of the factors as the share of mean absolute SHAP value on the test period. Recent symptom scores and flare-ups account for 68% of the attribution. Exposure factors together account for 11.7% (15.5% on validation), even though removing them does not lower the AUROC. The two results are consistent: SHAP describes what the trained model uses, and the ablation what it needs. Exposure is correlated with season and with the simulated symptom response, so a model trained without it reaches the same discrimination through those features. The additivity of equation (10) held to within 10^{-14} on every test prediction.

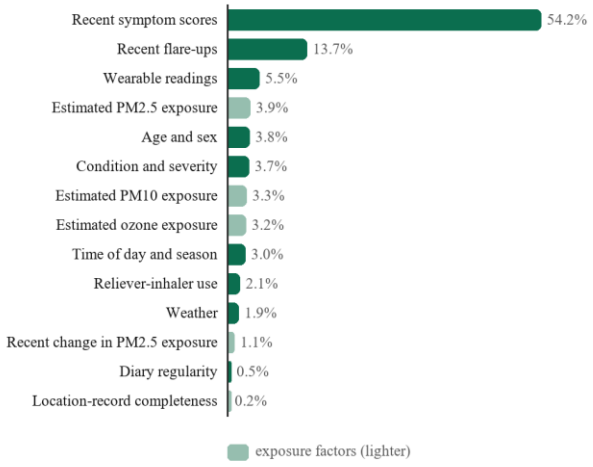


Fig. 4. Global importance: share of mean |SHAP| per factor, test, seen patients. Lighter bars are exposure factors.

XV. LIMITATIONS

All risk results come from a simulated cohort whose data-generating process we wrote; they show that the pipeline, calibration, explanations and alerts behave correctly and honestly, not that AeroGard predicts real exacerbations. Its effect sizes are assumptions, so the finding that exposure adds nothing is a property of this cohort and test season, not evidence about pollution and asthma. The exposure factors are not validated with personal sensors, NO_2 and SO_2 units remain unverified, and only one city and one monsoon test season were evaluated. Clinical use would require real, ethically collected outcomes and re-derived thresholds.

XVI. CONCLUSION

We built an end-to-end respiratory-alert system for Bengaluru, from validated ingestion to explained alerts. Careful validation mattered throughout: it exposed a CO unit error in the public feed, isolated suspect sensor spikes, caught a leakage bug in our own gap filling, and replaced a calibrator that reported impossible zero risks. The direct LightGBM forecaster reduced 24-hour $PM_{2.5}$ error to 5.78 µg/m³, 23% better than persistence. On a clearly labelled synthetic cohort the calibrated risk model reached an AUROC of 0.71 for seen and 0.72 for unseen patients, the symptom diary carried the signal, and exposure did not add measurable value. Explanations add up exactly to the probability shown, and alerts are raised once, traceably and without medication advice.

XVII. FUTURE SCOPE

Future work includes the Android application with push alerts and optional Health Connect data; a feasibility pilot with real patients under ethics approval; evaluation over a full year including winter, when exposure is higher; verification of NO_2 and SO_2 units against CPCB's own feed; comparison with sequence models such as LSTM [6]; forecast-driven weather inputs; and validation of the exposure factors with personal sensors.

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